

- 8 (a) State what is meant by a photon.

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..... [2]

- (b) Fig. 8.1 shows a tube in which X-rays are produced at a metal target.

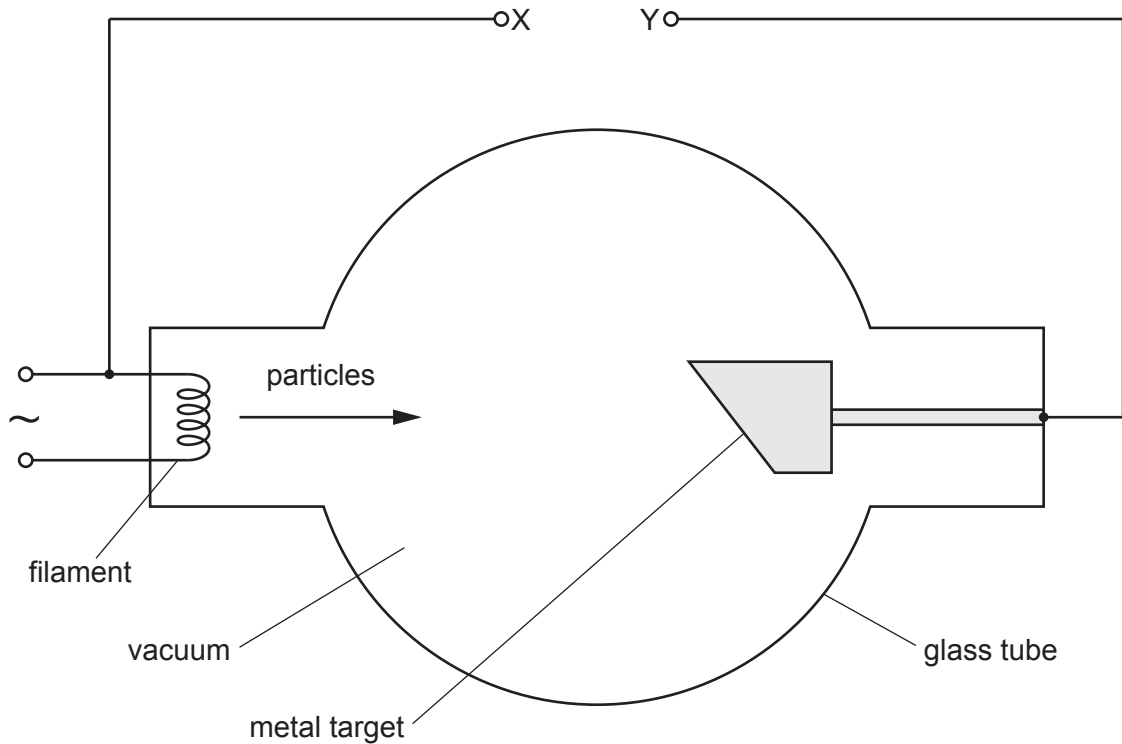


Fig. 8.1

Particles are accelerated from the filament to the target by a constant high voltage applied across the terminals X and Y.

- (i) State the name of the particles.

..... [1]

- (ii) On Fig. 8.1, use + and – signs to label terminals X and Y to indicate the polarity of the high voltage. [1]

(c) For an accelerating voltage of 32 kV in Fig. 8.1, determine:

(i) the maximum energy, in MeV, of an X-ray photon produced at the target

maximum photon energy = MeV [1]

(ii) the maximum momentum of an X-ray photon produced at the target

maximum photon momentum = N s [2]

(iii) the minimum wavelength of X-rays produced at the target.

minimum wavelength = m [3]

(d) Explain why X-rays can be used to produce images of internal body structures that have good contrast.

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..... [3]

[Total: 13]